



Banking Fraud Detection SQL Project



Project Overview

This project is a **Banking Fraud Detection Database System** built using **MySQL**.

It simulates real-world banking operations including:

- Customer Management
- Account Management
- Transaction Processing
- Fraud Detection & Risk Analysis

The database is fully relational with proper foreign key constraints and normalized schema design.



Database Structure

Total Tables: 4

1. Customers
2. Accounts
3. Transactions
4. Fraud_Alert

Relationships Implemented

- ✓ One-to-Many Relationship between Customer and Accounts
 - ✓ One-to-Many Relationship between Account and Transactions
 - ✓ One-to-Many Relationship between Transactions and Fraud Alerts
 - ✓ Referential Integrity maintained using Foreign Keys
 - ✓ Normalized Database Design (up to 3NF)
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Implemented SQL Business Scenarios

This project includes real-time banking and fraud analysis queries from basic to advanced level.

✓ Basic Level Queries

- Total number of customers
- Active vs Inactive accounts
- Total transaction amount
- Maximum and minimum transaction amount
- Top 5 transactions by amount
- Top 5 customers by account balance
- Customers with inactive savings accounts

✓ Intermediate Level Queries

- Customer-wise transaction history
- Fraud transactions with customer details
- Normal transactions report
- Average transaction amount by city
- High-value transaction detection
- Location-wise transaction analysis
- Top customers based on transaction amount

✓ Advanced Level Queries

- Customer Risk Score Classification
- Fraud-Prone Location Identification
- Banking Master Table Creation using JOINS
- Top Customers using Window Functions
- Top Customers using CTEs
- Fraud Pattern Analysis
- Risk Categorization using CASE Statements

SQL Concepts Demonstrated

- ✓ Database Design
- ✓ Primary Keys & Foreign Keys
- ✓ CRUD Operations
- ✓ INNER JOIN
- ✓ LEFT JOIN
- ✓ Aggregate Functions (COUNT, SUM, AVG, MAX, MIN)
- ✓ GROUP BY & HAVING

- ✓ CASE Statements
 - ✓ Common Table Expressions (CTE)
 - ✓ Window Functions (ROW_NUMBER)
 - ✓ Subqueries
 - ✓ Data Analysis Queries
 - ✓ Fraud Detection Logic
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Key Business Insights Generated

- Customers involved in suspicious transactions
 - High-risk customers based on fraud history
 - Locations generating maximum fraud alerts
 - High-value transaction monitoring
 - Average spending behavior by city
 - Active vs inactive account analysis
 - Fraud trend identification
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Project Files

- `banking_database.sql` → Database Schema & Table Creation
 - `sample_data.sql` → Sample Banking Data
 - `queries.sql` → SQL Analysis Queries
 - `er_diagram.png` → Database ER Diagram
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ER Diagram

(Add ER Diagram Screenshot Here)

⚙ Technologies Used

- MySQL
 - MySQL Workbench
 - Git & GitHub
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What This Project Demonstrates

- Strong understanding of Relational Database Design
 - Banking Domain Knowledge
 - Fraud Detection using SQL
 - Advanced Query Writing Skills
 - Data Analysis using SQL
 - Real-world Business Problem Solving
 - Hands-on Experience with Joins, CTEs, Window Functions, and Aggregations
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How to Run

Create Database

```
CREATE DATABASE Proj;  
USE Proj;
```

Run Table Creation Script

```
SOURCE banking_database.sql;
```

Insert Sample Data

```
SOURCE sample_data.sql;
```

Execute Analysis Queries

```
SOURCE queries.sql;
```

★ Project Highlights

- 30 Customers
- 30 Accounts
- 30 Transactions
- Fraud Detection Scenarios
- Risk Scoring System
- Banking Analytics Dashboard Ready Dataset
- Advanced SQL Concepts Implemented